# Multimodal Real Estate Valuation: Integrating Satellite Imagery with Tabular Data

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**Abstract**

Real estate valuation is traditionally addressed using structured housing attributes such as size, quality, and location. This project explores whether satellite imagery can provide complementary environmental context—such as greenery, road density, and waterfront proximity—to improve property price prediction. An end-to-end multimodal regression pipeline was implemented, combining tabular data with CNN-extracted visual embeddings from satellite images. While a tabular Random Forest baseline achieved strong performance ($R^2 \approx 0.88$), multimodal fusion experiments provided valuable insights into feature redundancy, environmental signal encoding, and the conditions under which visual data contributes to predictive modeling.

## 1 Introduction & Problem Statement

Property valuation is influenced not only by measurable attributes such as square footage or number of rooms, but also by qualitative environmental factors including neighborhood layout, green cover, and proximity to water bodies. Traditional tabular datasets often encode these factors only indirectly.

Satellite imagery provides a scalable and globally available source of environmental information. This project investigates whether incorporating satellite imagery into a structured real estate valuation pipeline improves predictive performance and, more importantly, whether it yields interpretable insights into how environmental context influences property prices.

The key research questions are:

- Can satellite imagery meaningfully complement structured housing attributes?
- Do visual features provide incremental predictive value beyond strong tabular proxies?
- What limitations arise when fusing high-dimensional visual features with tabular data?

## 2 Dataset Description

### 2.1 Tabular Data

The dataset consists of historical housing transactions from King County, USA, comprising approximately 16,000 training samples. The target variable is `price`, which was log-transformed during training to stabilize variance and reduce skewness.

Key predictors include:

- **sqft_living**: Interior living area.
- **grade**: Construction quality and architectural design.
- **sqft_living15**: Average living area of neighboring properties (neighborhood affluence proxy).
- **latitude & longitude**: Spatial clustering and image acquisition.

### 2.2 Satellite Image Data

One satellite image per property was fetched using the Mapbox Static Images API based on geographic coordinates. A fixed zoom level was selected to capture both the property footprint and immediate neighborhood context, including vegetation density, road networks, and nearby water bodies.

## 3 Data Acquisition Pipeline

An automated image-fetching pipeline was implemented in `data_fetcher.py`. Key engineering considerations included:

- Rate-limit compliance to avoid API throttling.
- One-to-one mapping between property IDs and image files.
- Graceful handling of missing or duplicate entries.

This design ensured reproducibility and scalability of the data acquisition process.

## 4 Exploratory Data Analysis

### 4.1 Price Distribution Analysis


Figure 1: Raw house price distribution.

The raw price distribution exhibits a strong right skew, with a long tail of high-value properties. This indicates heteroscedasticity and motivates transforming the target variable prior to regression.

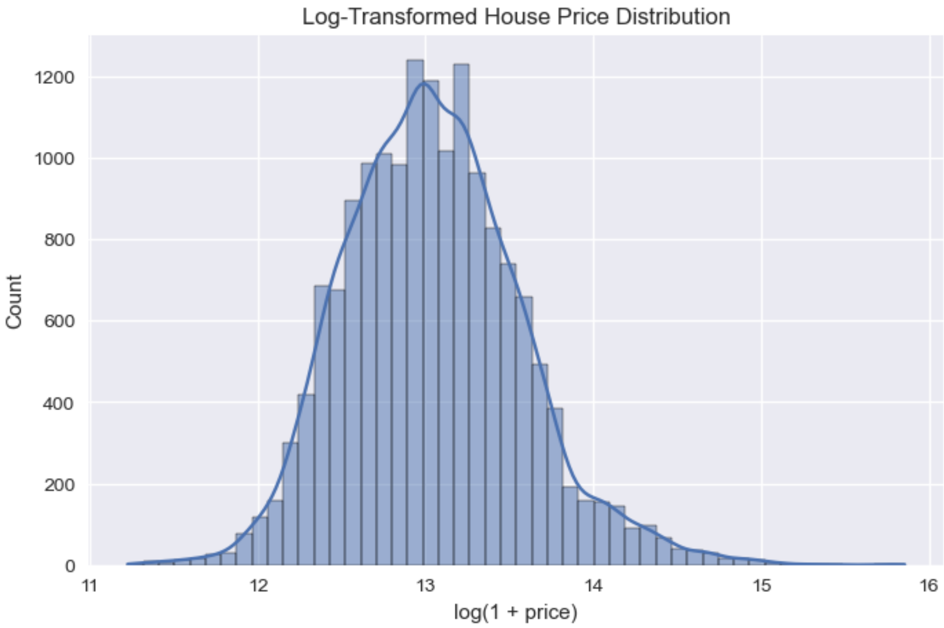


Figure 2: Log-transformed house price distribution.

After log transformation, the distribution becomes approximately normal. This improves model stability and aligns with assumptions of many regression algorithms.

### 4.2 Living Area vs Price


Figure 3: Living area versus log-transformed house price.

A strong positive relationship is observed between living area and price, with diminishing marginal returns at higher square footage. This confirms domain intuition and validates the importance of size-related features.

## 4.3 Spatial Price Patterns

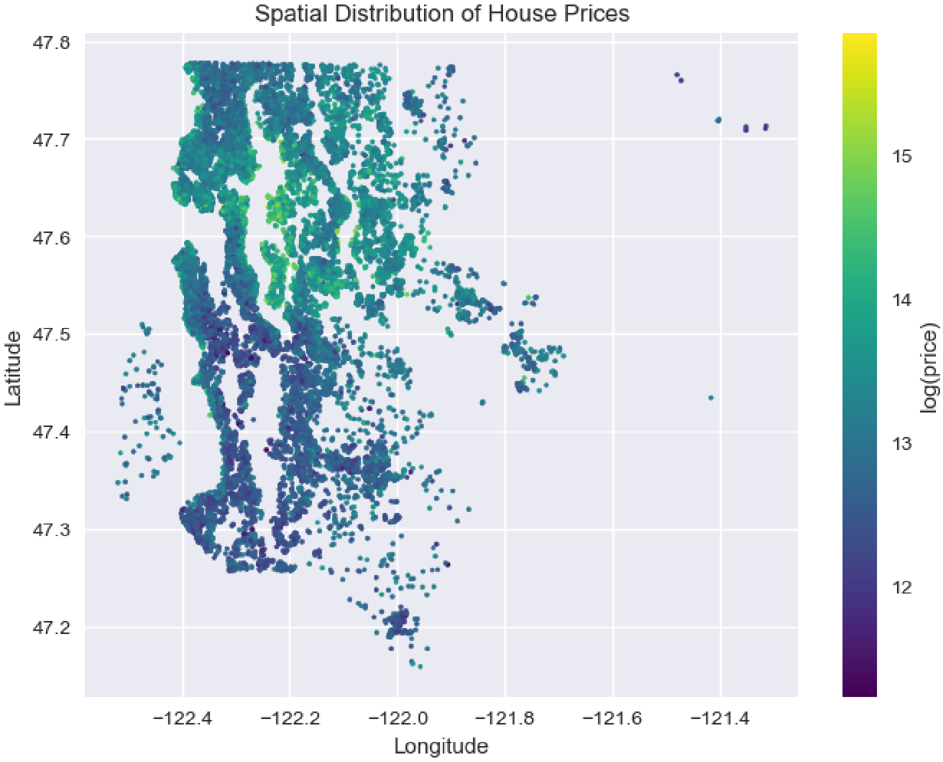


Figure 4: Spatial distribution of house prices.

Clear geographic clustering of high-value properties is evident. Premium properties are concentrated near waterfronts and central urban areas, suggesting that location-driven environmental context plays a significant role in valuation.

## 4.4 Feature Correlations

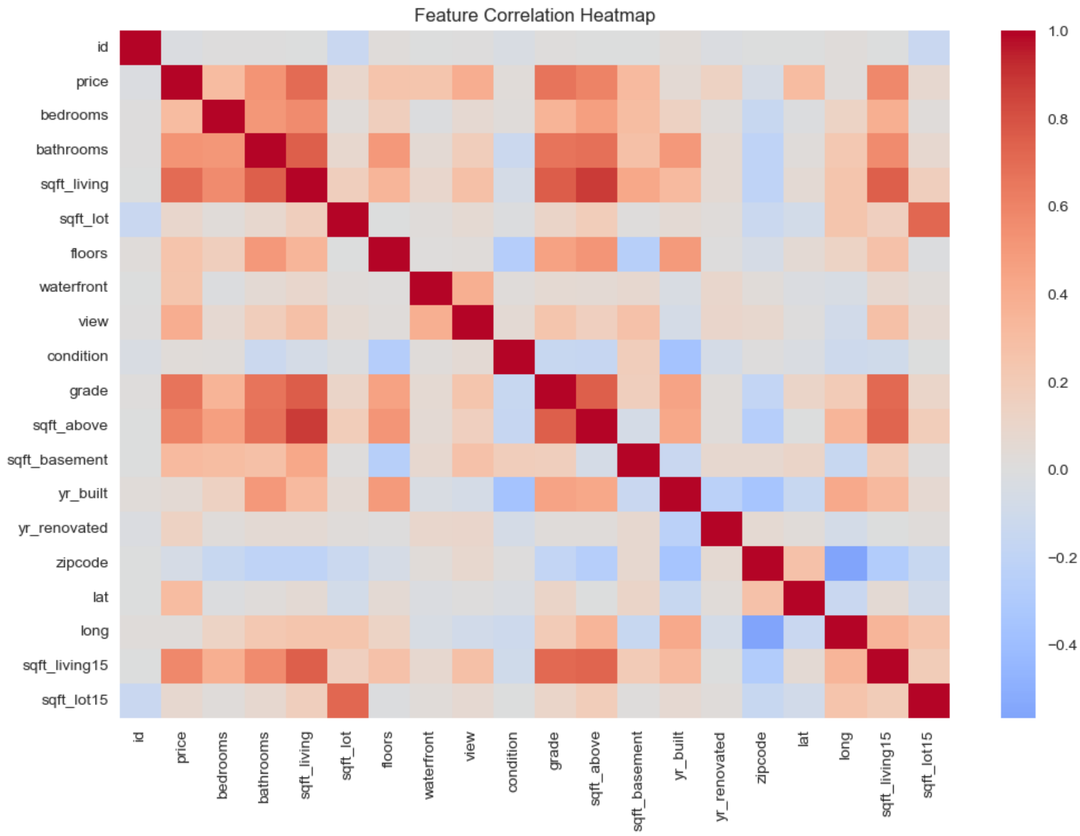


Figure 5: Feature correlation heatmap.

The correlation heatmap confirms that `sqft_living`, `grade`, and neighborhood-level features exhibit the strongest relationships with price. Many environmental factors captured visually are already partially encoded through these structured proxies.

# 5 Feature Engineering

## 5.1 Tabular Processing

Non-numeric columns were removed, duplicate property IDs were resolved, and all features were validated for completeness. The target variable was log-transformed to address skewness and stabilize variance.

## 5.2 Image Feature Extraction

A pretrained ResNet-18 CNN was used as a fixed feature extractor. Images were resized to $224 \times 224$ and normalized using ImageNet statistics. The final classification layer was removed, yielding a 512-dimensional embedding representing high-level visual semantics such as greenery, road structure, and water presence.

This approach avoids the need for large labeled satellite datasets while leveraging robust pretrained representations.

# 6 Modeling Approach

## 6.1 Baseline Model: Tabular Random Forest

A Random Forest Regressor was selected as the baseline due to its strong performance on structured data, ability to capture non-linear relationships, and robustness to feature scaling. This model establishes a strong benchmark against which multimodal approaches are evaluated.

## 6.2 Multimodal Fusion Strategies

Two fusion strategies were explored:

1. **Early Fusion with MLP**: Direct concatenation of tabular features and raw image embeddings followed by neural network regression. This approach suffered from overfitting due to high dimensionality.

2. **Tree-Based Fusion with PCA**: Image embeddings were reduced using PCA before fusion with tabular features and training a Random Forest. This approach improved stability while retaining most visual variance.

# 7 Results & Evaluation

| **Model** | **RMSE (log)** | $R^2$ |
|---|---|---|
| Tabular Random Forest | 0.1787 | 0.8843 |
| Multimodal RF (PCA Fused) | 0.1822 | 0.8727 |

Table 1: Performance comparison between baseline and multimodal models.

While the multimodal model performs competitively, the tabular baseline remains superior. This indicates that structured features already encode strong proxies for environmental quality, limiting the marginal benefit of satellite imagery in this dataset.

# 8 Explainability & Insights

Since the CNN was used as a frozen feature extractor rather than trained end-to-end for price prediction, gradient-based attribution methods such as Grad-CAM were not directly applicable. Interpretability was achieved through:

- Spatial clustering analysis of prices.
- Visual inspection of satellite imagery across price quantiles.
- Comparative model performance evaluation.

High-priced regions consistently exhibit higher vegetation density and waterfront proximity, confirming the semantic relevance of visual features.

# 9 Discussion, Limitations & Future Work

The dominance of tabular features suggests that satellite imagery provides limited incremental value when high-quality structured data is available. Additionally, fixed zoom levels may obscure fine-grained property details.

Future work includes fine-tuning CNNs on geospatial data, incorporating multi-scale imagery, and applying this pipeline to regions with sparse or unreliable tabular records.

## 10 Conclusion

This project implemented a complete multimodal real estate valuation pipeline integrating satellite imagery with tabular housing data. While the tabular Random Forest remained the strongest predictor, the multimodal framework provides valuable insights into environmental context modeling and demonstrates a scalable approach for real-world deployment.

## 11 Prediction Distribution Validation

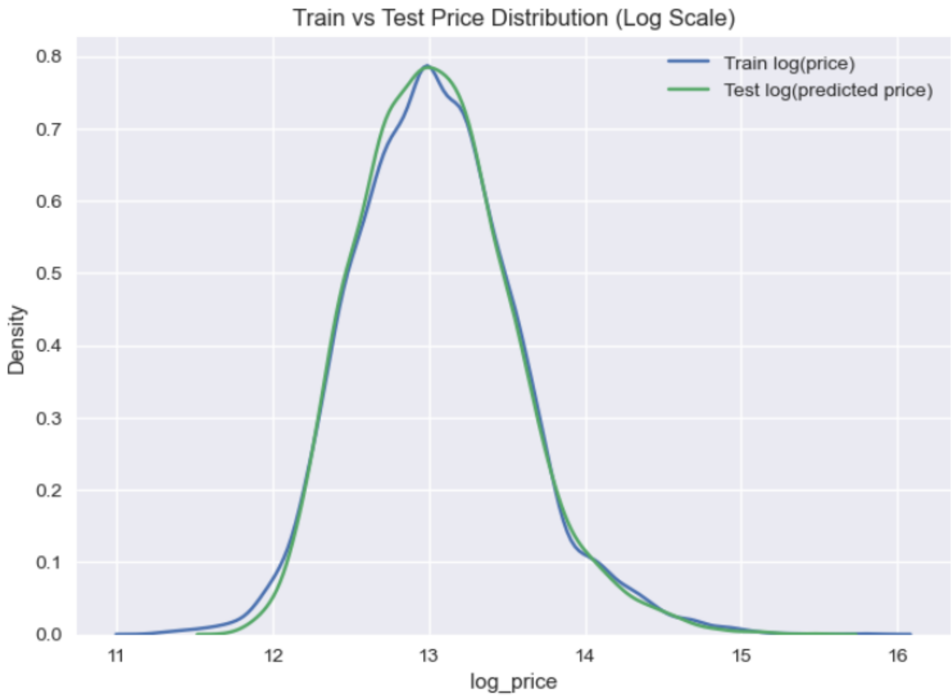


Figure 6: Comparison of log-transformed training prices and predicted test prices.

Figure 6 compares the distribution of log-transformed prices in the training data with the distribution of predicted prices on the test set. The strong overlap between the two curves indicates that the model's predictions are well-calibrated and lie within a realistic price range.

This alignment suggests that the model has not suffered from severe overfitting or distributional shift, and that the learned relationships generalize effectively to unseen data. While this analysis does not replace ground-truth evaluation on the test set, it serves as an important sanity check validating the plausibility and stability of the final predictions.